SCHOOL OF TECHNOLOGY

FINAL YEAR PROJECT II - SYSTEMS IMPLEMENTATION TRACK

STAFF TRAINING MATERIAL

USER TRAINING MANUAL

Operating Guide for the MedPlus Pharmacy POS & Inventory Management System (2030 Glassmorphic Edition)

INTENDED AUDIENCE:

Pharmacists, Cashiers, Stock Managers, and Administrators

SYSTEM VERSION:

v1.0.0 • Whole Integer Ksh Pricing

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> **IMPORTANT SAFETY AND COMPLIANCE NOTICE**
>
> This manual documents the software operations of the MedPlus Pharmacy POS system. The system enforces clinical prescription validation and payment reporting standards. Users must ensure that restricted medicines (Rx category) are only dispensed after verifying a valid doctor's prescription. Patient records stored within the database are subject to regional medical confidentiality laws. Unauthorized sharing of staff credentials or bypasses of dynamic Role-Based Access Control matrix fields is strictly prohibited.

All patient identifiers, prescriptions, and supplier databases used in this guide are illustrative. Safaricom Daraja API configurations and M-Pesa STK callbacks must only be connected using certified merchant credentials.

Preface

This training manual explains how to operate the **MedPlus Pharmacy POS & Inventory Management System** in plain language. It guides staff through daily sales, stock management, procurement, and administrative duties.

Target Audience Definitions

- **Pharmacists & Cashiers:** Staff handling front-desk sales, validating prescriptions, scanning barcodes, receiving Cash/M-Pesa payments, and issuing customer receipts.
- **Stock & Inventory Managers:** Staff tracking stock levels, receiving supply deliveries, and submitting purchase orders to vendors.
- **System Administrators:** Technicians managing user creation, dynamic Role-Based Access Control (RBAC) matrices, password resets, and database audits.

System Documentation Directory

This guide is part of a comprehensive documentation suite. For developer-focused specifications, please refer to:

1. **System Requirements Specification (SRS):** Core requirements and API boundaries.
2. **System Design Document (SDS):** Database schemas, code structure, and route designs.
3. **MedPlus QA Test Plan:** Verification scripts, edge case handling, and integration tests.

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Module 1: Access & System Authentication

The MedPlus Pharmacy POS system is accessed through the web browser. The system is built with a 2030 dark-glassmorphism visual theme designed to reduce eye strain during long shifts.

Step 1: Accessing the Login Portal

Open the Google Chrome browser on your terminal and navigate to the system URL provided by your administrator (typically `http://localhost:8080` for local containers, or the production URL).

Step 2: Entering User Credentials

1. Enter your registered **Username**.
2. Enter your secure **Password**.
3. Click the **Sign In** button to execute credentials verification.

The backend Go server verifies your password against the stored bcrypt hash. Upon validation, the server issues a secure JSON Web Token (JWT) that is saved locally in your browser's session.

Step 3: Accessing the Dashboard

Once logged in, the system runs the `bootstrapAuthenticatedApp()` function, retrieving your profile, active role, and permission matrix to load only the modules you are authorized to see.

> **Session Invalidation (Logout)**
> To secure patient data, always click the **Logout** button in the sidebar before leaving your workstation. This deletes the saved JWT token from your browser and returns you to the login screen.

Module 2: Point of Sale - Cart & Checkout (Part 1)

The Point of Sale (POS) interface is the primary workspace for pharmacists and cashiers. It is designed to process product checkouts quickly and validate prescriptions in real time.

Step 1: Finding Products

Open the **POS** tab from the sidebar. You can search the product catalog using three methods:

- **Search Bar:** Type the name of the medicine (e.g., "Amoxicillin") in the text box. The list filters dynamically as you type.
- **Category Filter:** Click the category tabs (e.g. *Over the Counter (OTC)*, *Prescription (Rx)*, *Vitamins*) to view specific product catalogs.
- **Barcode Scanning:** Scan the item's barcode. The system matches the barcode and automatically adds the item to the cart.

Step 2: Managing the Shopping Cart

1. Click a product card to add the item to the cart.
2. To adjust quantities, click the **+** or **-** buttons inside the cart widget. The system checks stock levels in real time and blocks entries that exceed available inventory.
3. The system displays prices in **whole integer Kenyan Shillings (Ksh)** without cents or decimals.

Step 3: Prescription (Rx) Verification Workflow

If an item is marked as **Prescription-Only (Rx)**, adding it to the cart triggers the verification routine:

> **Prescription Validation Steps**
>
> 1. A modal prompt displays: **"Prescription ID Required"**.
> 2. Enter the prescription number (issued by the doctor).
> 3. Click **Verify**. The frontend checks the prescription ID via the `/api/prescriptions/{id}` endpoint.
> 4. If valid and active, the modal closes, the cart records the prescription ID, and a green **"Verified"** badge appears on the item.
> 5. If the prescription ID is invalid, expired, or doesn't exist, the system blocks the checkout.

Note: Over-the-Counter (OTC) items do not trigger this verification step.

Module 2: Point of Sale - Checkout & Receipts (Part 2)

Once all items are in the cart and prescriptions are verified, you can complete the sale. The system supports Cash, Card, and M-Pesa payments.

Step 4: Selecting a Customer

By default, the cart uses a generic *"Walk-in Customer"*. To link a customer, search for their name or phone number in the customer search box. Linking a customer updates their profile with transaction metrics and loyalty records.

Step 5: Processing the Payment

Choose the customer's payment method from the checkout interface:

Option A: Cash Payment

1. Select **Cash**.
2. Enter the amount received from the customer. The system calculates and displays the change due in Ksh.
3. Click **Complete Checkout**.

Option B: M-Pesa STK Push Integration

The system integrates with Safaricom's Daraja API to trigger a direct STK push to the customer's phone:

1. Select **M-Pesa** as the payment method.
2. Enter the customer's active phone number (format: `07XXXXXXXX` or `254XXXXXXXXX`).
3. Click **Send STK Push**. The screen displays a loading spinner showing the status as `Pending Customer PIN`.
4. The customer receives a popup on their phone requesting their M-Pesa PIN for the exact Ksh checkout total.
5. Once the customer enters their PIN, Safaricom sends a callback webhook to the Go backend, updating the payment audit status.
6. The screen updates with a green **Payment Successful** checkmark, completing the sale.

Option C: Card Payment

Select **Card**, swipe/tap the card on the external merchant terminal, and click **Complete Checkout**.

Step 6: Receipt Printing

Upon successful checkout, the inventory balances update immediately. A modal pops up with the receipt preview. Click **Print Receipt** to send the print job to the thermal printer.

Module 3: Inventory Catalog & Expiry Controls

The Inventory module allows stock managers and pharmacists to monitor stock availability, check expiry dates, and update product details.

Viewing Stock and Expiry Alerts

Open the **Inventory** tab. The dashboard widgets display:

- **Low Stock Indicators:** Highlighted items where current stock is below the defined reorder alert level.
- **Expiring Medicines Alert:** Lists items expiring within 180 days, allowing staff to run promotional sales or return stock before expiry.

Adding New Products

1. Navigate to the **Inventory Manager** panel (requires Admin or authorized Pharmacist role).
2. Click **Add Product**.
3. Enter the product name, SKU/Barcode, category (OTC/Rx), price in Ksh, reorder alert level, and expiry date.
4. Click **Save Product**. The item is saved to the PostgreSQL catalog.

Deactivating Products

If a medicine is discontinued, do not delete it, as this would break historical sales reports. Instead, open the product details and click **Deactivate**. This soft-deactivates the item so it is hidden from checkouts while preserving database history.

Module 4: Customer Registration

Saving patient profiles allows for faster checkout processing and historical prescription checks.

Step 1: Adding a Patient Profile

1. Navigate to the **Customers** tab.
2. Click **Add Customer**.
3. Enter the patient's full name, phone number, and email.
4. Click **Register Customer**.

Step 2: Checking Customer History

Click on any customer name to view their profile details, past transaction dates, prescription histories, and loyalty account metrics.

Module 5: Supply Chain & Procurement

The Supply Chain module handles supplier databases, purchase orders (POs), and restocking workflows. It connects procurement records to active inventory updates.

Step 1: Registering a Supplier

1. Open the **Supply Chain** tab and select the **Suppliers** panel.
2. Click **Register Supplier**.
3. Enter the supplier business name, contact person, phone number, email, and address.
4. Click **Save**. The supplier record is added to the system database.

Step 2: Creating a Purchase Order (PO)

1. Navigate to the **Purchase Orders** panel.
2. Click **Create Purchase Order**.
3. Select the registered supplier from the dropdown list.
4. Select products and input quantities to order. The system calculates the estimated restock cost rounded to the nearest whole integer Kenyan Shilling (Ksh), applying a 30% merchant wholesale discount (`Math.round(price * 0.7)`).
5. Click **Create PO**. The PO is created in `Draft` status.
6. Review the PO details and click **Submit Purchase Order**. The status updates to `Submitted`, locking the record.

Step 3: Receiving Stock & Inventory Updates

When the supplier delivers the ordered products:

> **Stock Ingestion Process**
>
> 1. Open the pending purchase order.
> 2. Inspect the delivered items against the PO quantities.
> 3. Click **Receive Stock**.
> 4. The backend transitions the PO status to `Received` and automatically updates the product inventory levels.
> 5. The PO is locked to prevent duplicate ingestion. An audit entry logs the receiving pharmacist's details.

Module 6: Dashboard Analytics & Reports

The Reports module compiles database transactions into live metrics for managers and owners.

Dashboard Feed Interpretation

- **Financial Metrics:** Live widgets tracking Today's Revenue, Total Completed Sales, and Active Prescription count.
- **Recent Transactions:** A list of the latest 5 checkout records, showing the cashier, total value, and payment method.
- **Expiring Alerts:** An alert widget displaying stock items expiring soon.

Generating Reports

Open the **Reports** tab. Choose from the available options:

- **Revenue Reports:** Summary of cash flow, card metrics, and M-Pesa totals.
- **Category Performance:** Graphic table showing units sold by category (OTC/Rx).
- **Inventory Valuation:** Total value of stock in the store, calculated by cost metrics.

Module 7: System Administration & Dynamic RBAC

Administrators use this panel to configure staff roles, set system keys, and review database schemas.

Dynamic Role-Based Access Control (RBAC)

MedPlus features dynamic permissions. To adjust permissions:

1. Open the **RBAC** panel under settings.
2. Select the role to modify (e.g. *Pharmacist*).
3. Toggle permission checkboxes (e.g. `view:database_overview`).
4. Click **Update Role Permissions**. The changes apply immediately without restarting the application.

Admin Database Schema Overview

To inspect table sizes and schema column types:

1. Open the **Database Overview** tab.
2. The system prompts for security re-verification. **Enter your admin password/PIN**.
3. Click **Verify Access**. This issues a 10-minute access token (`X-Database-Access-Token`).
4. The page lists the 11 database tables, row counts, and sizes in KB. Click any table name to open the schema modal displaying column names, SQL types, and nullable flags.

Reference Materials

Use these reference charts for keyboard shortcuts and to resolve system errors.

Keyboard Shortcuts

Shortcut Key	Action Triggered	Active View / Scope
`F1`	Opens POS checkout page.	Global Application Context
`F2`	Focuses product search box.	POS Cart Interface
`F4`	Opens Customer Registration modal.	POS Cart Interface
`F9`	Triggers cash checkout submission.	POS Cart Interface
`Esc`	Closes active modal/receipt dialog.	Modal Overlay Context

System Error Codes & Meaning

Error Displayed	System Cause	Corrective Action Required
`[API 401] Unauthorized Token`	JWT authentication token has expired or is invalid.	Log out and log back in to renew your session token.
`[API 403] Insufficient Permissions`	The active user role lacks permissions to access the module.	Contact system administrator to adjust the RBAC matrix.
`[POS 601] Rx Code Required`	Cart contains prescription-only items without a linked verification ID.	Prompt user for prescription number and submit validation modal.
`[Daraja 504] Callback Timeout`	M-Pesa API STK push did not receive a customer response in 60s.	Verify customer phone number, check mobile connectivity, and trigger push again.
`[DB 500] Connection Refused`	Go backend cannot connect to the PostgreSQL container.	Restart the database container or verify Docker network settings.

Frequently Asked Questions (FAQ)

Q: Why does the system round all transactions to whole Kenyan Shillings?
A: MedPlus uses whole integer Kenyan Shilling (Ksh) pricing to reflect local transactions. Cents have been removed from the database migrations and checkout pages.

Q: Can pharmacists modify the database structures?
A: No. Pharmacists handle inventory, checkouts, and customer databases. Modifying table structures requires the Admin role and password re-verification on the Database Overview screen.

Q: What happens if the M-Pesa API STK push fails to display?
A: Ensure the phone number uses the format `07XXXXXXXX` or `254XXXXXXXXX`. Check that the internet connection is active and the Safaricom Daraja credentials are valid.

Glossary of Terms

API (Application Programming Interface): The communication bridge between the browser and the Go server.

Backend: The Go server that handles application logic, verifies JWT sessions, and queries the database.

Daraja API: Safaricom's developer portal used to trigger M-Pesa STK payment pushes.

FIFO (First In, First Out): Stock logic ensuring older batches are processed first.

Frontend: The user interface accessed through the web browser.

JWT (JSON Web Token): A secure, encrypted session key used to verify user sessions.

RBAC (Role-Based Access Control): Security access governance determined by permission matrices.

Rx: Standard shorthand designating prescription-only medications.

STK Push (Simulated Transaction Keyboard): A mobile payment trigger prompting users for their PIN on their phone.